VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum- 590014, Karnataka.

LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Rayhan Sadat (1BM23CS264),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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		• Create a keyspace by name Employee
		• Create a column family by name
		  ○ Employee-Info with attributes
		  ○ Emp_Id Primary Key, Emp_Name, Designation,
		  ○ Date_of_Joining, Salary, Dept_Name
		• Insert the values into the table in batch
		• Update Employee name and Department of EmpId 121
		• Sort the details of Employee records based on salary
		• Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
		• Update the altered table to add project names.
		• Create a TTL of 15 seconds to display the values of Employees.
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3	16-02-2026	Perform the following DB operations using Cassandra.
		• Create a keyspace by name Library
		• Create a column family by name Library-Info with attributes
		  ○ Stud_Id Primary Key,
		  ○ Counter_value of type Counter,
		  ○ Stud_Name, Book-Name, Book-Id,
		  ○ Date_of_issue
		• Insert the values into the table in batch
		• Display the details of the table created and increase the value of the counter
		• Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
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Github Link: https://github.com/RayhanSadat26/BDA-LAB

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Command with output:

```
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.insertOne({_id: 1, StudName: "MichelleJacintha",
 Grade: "VII", Hobbies: "InternetSurfing"})
{ acknowledged: true, insertedId: 1 }
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.updateOne({_id: 2, StudName: 'AryanDavid', Grade
: 'VII'}, {$set: {Hobbies: "Skating"}}, {upsert: true})
{
  acknowledged: true,
  insertedId: 2,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 1
}
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.insertMany([{_id: 3, StudName: 'Charan', Grade:
'VII'}, {_id: 4, StudName: 'Vibinn', Grade: 'VII'}])
{ acknowledged: true, insertedIds: { '0': 3, '1': 4 } }
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.updateOne({_id: 3}, {$set: {Hobbies: 'Drawing'}}
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
Atlas atlas-ru5tdz-shard-0 [primary] test> use myDB
switched to db myDB
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db
myDB
Atlas atlas-ru5tdz-shard-0 [primary] myDB> show dbs
admin  232.00 KiB
local   18.01 GiB
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.show()
TypeError: db.show is not a function
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.show
myDB.show
Atlas atlas-ru5tdz-shard-0 [primary] myDB> show collections

Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.createCollection('Student')
{ ok: 1 }
Atlas atlas-ru5tdz-shard-0 [primary] myDB> show collections
Student
```

```
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.updateOne({_id: 2, StudName: 'Charan', Grade: 'V
II'}, {$set: {Hobbies: 'Drawing'}}, {upsert: false})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.updateMany({_id: 4}, {$set: {Hobbies: 'Drawing'}
})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.deleteOne({_id: 1})
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.deleteMany({Hobbies: 'Drawing'})
{ acknowledged: true, deletedCount: 2 }
Atlas atlas-ru5tdz-shard-0 [primary] myDB>
```

```
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.find()
[
  {
    _id: 1,
    StudName: 'MichelleJacintha',
    Grade: 'VII',
    Hobbies: 'InternetSurfing'
  },
  { _id: 2, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' },
  { _id: 3, StudName: 'Charan', Grade: 'VII', Hobbies: 'Drawing' },
  { _id: 4, StudName: 'Vibinn', Grade: 'VII', Hobbies: 'Drawing' }
]
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.find({StudName: 'DavidAryan'})

Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.find({StudName: 'AryanDavid'})
[
  { _id: 2, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' }
]
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.find({StudName: 'AryanDavid'},{_id: 0})
[ { Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' } ]
Atlas atlas-ru5tdz-shard-0 [primary] myDB> db.Student.find({Grade: {$eq :'VII'}})
[
  {
    _id: 1,
    StudName: 'MichelleJacintha',
    Grade: 'VII',
    Hobbies: 'InternetSurfing'
  },
  { _id: 2, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' },
  { _id: 3, StudName: 'Charan', Grade: 'VII', Hobbies: 'Drawing' },
  { _id: 4, StudName: 'Vibinn', Grade: 'VII', Hobbies: 'Drawing' }
]
```

PROGRAM - 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

Command with output :

```
MyDataBase> use MyDataBase
already on db MyDataBase
MyDataBase> show collections
Customers
NewStudent
Student
MyDataBase> db.Student.find();
[
  {
    _id: 1,
    studName: 'Michellejacintha',
    Grade: 'VII',
    Hobbies: 'InternetSurfing'
  },
  { _id: 3, Grade: 'VII', StudName: 'AryanDavid', Hobbies: 'Skating' },
  { _id: 2, Grade: 'VIII', StudName: 'Ram', Hobbies: 'Learning' }
]
```

```
test> use MyDataBase
switched to db MyDataBase
MyDataBase> show collections
NewStudent
NewStudent2
Student
MyDataBase> db.NewStudent2.drop();
true
MyDataBase> db.createCollection("Customers");
{ ok: 1 }
MyDataBase> db.Customers.insertMany([{cust_id:1,Balance:200, Type:"S"},]);
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d00571207666297fa3b81a') }
}
MyDataBase> db.Customers.insert({cust_id:1,Balance:1000, Type:"Z"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0058f207666297fa3b81b') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:100, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d0059c207666297fa3b81c') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:1000, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005a5207666297fa3b81d') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:500, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ad207666297fa3b81e') }
}
MyDataBase> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005b2207666297fa3b81f') }
}
MyDataBase> db.Customers.insert({cust_id:3,Balance:500, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67d005ba207666297fa3b820') }
}
```

```
MyDataBase> db.Customers.aggregate([
...   {
...     $group: {
...       _id: "$cust_id",              // Group by cust_id
...       minAccBal: { $min: "$Balance" }, // Find the minimum Balance
...       maxAccBal: { $max: "$Balance" }  // Find the maximum Balance
...     }
...   }
... ]);
[
  { _id: 3, minAccBal: 500, maxAccBal: 500 },
  { _id: 2, minAccBal: 50, maxAccBal: 1000 },
  { _id: 1, minAccBal: 200, maxAccBal: 1000 }
]
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   { $group: { _id: "$cust_id", TotAccBal: { $sum: "$Balance" } } },
...   { $match: { TotAccBal: { $gt: 1200 } } }
... ]);
```

```
MyDataBase> db.Customers.aggregate([
...   { $match: { Type: "Z" } },
...   {
...     $group: {
...       _id: "$cust_id",
...       TotAccBal: { $sum: "$Balance" }
...     }
...   },
...   {
...     $match: {
...       TotAccBal: { $gt: 200 }
...     }
...   }
... ]);
[ { _id: 3, TotAccBal: 500 }, { _id: 1, TotAccBal: 1000 } ]
```

```
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoimport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=csv --file /home/bmscecse/Desktop/135.txt --headerline
2025-03-11T14:55:05.192+0530    connected to: mongodb://localhost/
2025-03-11T14:55:05.360+0530    3 document(s) imported successfully. 0 document(s) failed to import.
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --file /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:24.438+0530    error parsing command line options: unknown option "file"
2025-03-11T14:55:24.438+0530    try 'mongoexport --help' for more information
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ mongoexport --host localhost --db MyDataBase --coll
ection NewStudent2 --type=json --out /home/bmscecse/Desktop/135.txt
2025-03-11T14:55:32.771+0530    connected to: mongodb://localhost/
2025-03-11T14:55:32.780+0530    exported 3 records
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Employee
b) Create a column family by name
- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

Command with output:

```
cqlsh> CREATE KEYSPACE IF NOT EXISTS Employee
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Employee;
cqlsh:employee> CREATE TABLE IF NOT EXISTS Employee_Info (
            ...   Emp_Id INT PRIMARY KEY,
            ...   Emp_Name TEXT,
            ...   Designation TEXT,
            ...   Date_of_Joining DATE,
            ...   Salary DOUBLE,
            ...   Dept_Name TEXT
            ... );
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (121, 'John Doe', 'Manager', '2018-01-01', 90000, 'HR');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (122, 'Alice Smith', 'Developer', '2019-05-21', 75000, 'IT');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (123, 'Rahul Roy', 'Analyst', '2020-07-15', 65000, 'IT');
            ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info
            ... SET Emp_Name = 'John Smith', Dept_Name = 'Finance'
            ... WHERE Emp_Id = 121;
cqlsh:employee> select * from Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | salary
--------+-----------------+-----------+-------------+-------------+--------
    123 |      2020-07-15 |        IT |     Analyst |   Rahul Roy |  65000
    122 |      2019-05-21 |        IT |   Developer | Alice Smith |  75000
    121 |      2018-01-01 |   Finance |     Manager |  John Smith |  90000

(3 rows)
```

```
(3 rows)
cqlsh:employee> CREATE TABLE IF NOT EXISTS Employee_By_Dept (
            ...   Dept_Name TEXT,
            ...   Salary DOUBLE,
            ...   Emp_Id INT,
            ...   Emp_Name TEXT,
            ...   Designation TEXT,
            ...   Date_of_Joining DATE,
            ...   PRIMARY KEY (Dept_Name, Salary, Emp_Id)
            ... ) WITH CLUSTERING ORDER BY (Salary DESC, Emp_Id ASC);
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_By_Dept (Dept_Name, Salary, Emp_Id, Emp_Name, Designation, Date_of_Joining)
            ... VALUES ('HR', 90000, 121, 'John Smith', 'Manager', '2018-01-01');
            ...
            ... INSERT INTO Employee_By_Dept (Dept_Name, Salary, Emp_Id, Emp_Name, Designation, Date_of_Joining)
            ... VALUES ('IT', 75000, 122, 'Alice Smith', 'Developer', '2019-05-21');
            ...
            ... INSERT INTO Employee_By_Dept (Dept_Name, Salary, Emp_Id, Emp_Name, Designation, Date_of_Joining)
            ... VALUES ('IT', 65000, 123, 'Rahul Roy', 'Analyst', '2020-07-15');
            ... APPLY BATCH;
cqlsh:employee> SELECT * FROM Employee_By_Dept WHERE Dept_Name = 'IT';

 dept_name | salary | emp_id | date_of_joining | designation | emp_name
-----------+--------+--------+-----------------+-------------+-------------
        IT |  75000 |    122 |      2019-05-21 |   Developer | Alice Smith
        IT |  65000 |    123 |      2020-07-15 |     Analyst |   Rahul Roy

(2 rows)
cqlsh:employee> ALTER TABLE Employee_Info ADD Projects SET<TEXT>;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ERP System', 'HR Portal'} WHERE Emp_Id = 121;
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (124, 'Sneha Kapoor', 'Tester', '2023-03-10', 55000, 'QA') USING TTL 15;
cqlsh:employee> select * from Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                    | salary
--------+-----------------+-----------+-------------+-------------+-----------------------------+--------
    123 |      2020-07-15 |        IT |     Analyst |   Rahul Roy |                        null |  65000
    122 |      2019-05-21 |        IT |   Developer | Alice Smith |                        null |  75000
    121 |      2018-01-01 |   Finance |     Manager |  John Smith | {'ERP System', 'HR Portal'} |  90000

(3 rows)
```

PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes
- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

Command with output:

```
cqlsh:employee> CREATE KEYSPACE IF NOT EXISTS Library
           ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh:employee> USE Library;
cqlsh:library> CREATE TABLE IF NOT EXISTS Library_Info (
           ...   Stud_Id INT PRIMARY KEY,
           ...   Stud_Name TEXT,
           ...   Book_Name TEXT,
           ...   Book_Id TEXT,
           ...   Date_of_issue DATE
           ... );
cqlsh:library> CREATE TABLE IF NOT EXISTS Book_Counter (
           ...   Stud_Id INT,
           ...   Book_Name TEXT,
           ...   Counter_value COUNTER,
           ...   PRIMARY KEY ((Stud_Id), Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Anjali Rao', 'BDA', 'B101', '2024-10-01');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Karthik N', 'AI', 'B102', '2024-11-11');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> UPDATE Book_Counter SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library> SELECT * FROM Book_Counter WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             4

(1 rows)
```

```
cqlsh:students> DESCRIBE TABLE Students_Info;

CREATE TABLE students.students_info (
    roll_no int PRIMARY KEY,
    dateofjoining timestamp,
    last_exam_percent double,
    studname text
) WITH additional_write_policy = '99p'
    AND bloom_filter_fp_chance = 0.01
    AND caching = {'keys': 'ALL', 'rows_per_partition': 'NONE'}
    AND cdc = false
    AND comment = ''
    AND compaction = {'class': 'org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy', 'max_threshold': '32', 'min_threshold': '4'}
    AND compression = {'chunk_length_in_kb': '16', 'class': 'org.apache.cassandra.io.compress.LZ4Compressor'}
    AND memtable = 'default'
    AND crc_check_chance = 1.0
    AND default_time_to_live = 0
    AND extensions = {}
    AND gc_grace_seconds = 864000
    AND max_index_interval = 2048
    AND memtable_flush_period_in_ms = 0
    AND min_index_interval = 128
    AND read_repair = 'BLOCKING'
    AND speculative_retry = '99p';
cqlsh:students> BEGIN BATCH
            ... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
            ... VALUES (1, 'Asha', '2012-03-12', 79.9);
            ... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
            ... VALUES (2, 'Kiran', '2012-03-12', 89.9);
            ... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
            ... VALUES (3, 'Shanthi', '2012-03-12', 90.9);
            ... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
            ... VALUES (4, 'Smith', '2012-03-12', 67.9);
            ... INSERT INTO Students_Info (Roll_No, StudName, DateOfJoining, Last_Exam_Percent)
            ... VALUES (5, 'Rohan', '2012-03-12', 56.9);
            ... APPLY BATCH;
cqlsh:students> SELECT * FROM Students_Info;

 roll_no | dateofjoining                   | last_exam_percent | studname
---------+---------------------------------+-------------------+----------
       5 | 2012-03-11 18:30:00.000000+0000 |              56.9 |    Rohan
       1 | 2012-03-11 18:30:00.000000+0000 |              79.9 |     Asha
       2 | 2012-03-11 18:30:00.000000+0000 |              89.9 |    Kiran
       4 | 2012-03-11 18:30:00.000000+0000 |              67.9 |    Smith
       3 | 2012-03-11 18:30:00.000000+0000 |              90.9 |  Shanthi

(5 rows)
```

```
cqlsh> CREATE KEYSPACE Students WITH REPLICATION =
   ... {'class': 'SimpleStrategy', 'replication_factor': '1'};
cqlsh>
cqlsh> USE Students;
cqlsh:students> DESCRIBE KEYSPACES;

companies  library      products    system              system_traces
company    pro          productss   system_auth         system_views
employe    prod         productsss  system_distributed  system_virtual_schema
employee   productname  students    system_schema

cqlsh:students> CREATE TABLE Students_Info (
            ...     Roll_No int PRIMARY KEY,
            ...     StudName text,
            ...     DateOfJoining timestamp,
            ...     last_exam_Percent double
            ... );
cqlsh:students> SELECT * FROM system.schema_keyspaces;
InvalidRequest: Error from server: code=2200 [Invalid query] message="table schema_keyspaces does not exist"
cqlsh:students>
cqlsh:students> SELECT * FROM system_schema.keyspaces;

 keyspace_name      | durable_writes | replication
--------------------+----------------+-------------------------------------------------------------------------------------
          companies |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
        system_auth |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
      system_schema |           True |                                {'class': 'org.apache.cassandra.locator.LocalStrategy'}
            library |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
           products |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
 system_distributed |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '3'}
             system |           True |                                {'class': 'org.apache.cassandra.locator.LocalStrategy'}
          productsss |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
               prod |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
                pro |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
      system_traces |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '2'}
           students |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
            company |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
           employee |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
        productname |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
            employe |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}
           productss |           True | {'class': 'org.apache.cassandra.locator.SimpleStrategy', 'replication_factor': '1'}

(17 rows)
cqlsh:students> DESCRIBE TABLES;

students_info
```

```
cqlsh:students> SELECT * FROM Students_Info WHERE Roll_No IN (1,2,3);

 roll_no | dateofjoining                   | last_exam_percent | studname
---------+---------------------------------+-------------------+----------
       1 | 2012-03-11 18:30:00.000000+0000 |              79.9 |     Asha
       2 | 2012-03-11 18:30:00.000000+0000 |              89.9 |    Kiran
       3 | 2012-03-11 18:30:00.000000+0000 |              90.9 |  Shanthi

(3 rows)
cqlsh:students> CREATE INDEX ON Students_Info (StudName);
cqlsh:students> SELECT * FROM Students_Info WHERE StudName = 'Asha';

 roll_no | dateofjoining                   | last_exam_percent | studname
---------+---------------------------------+-------------------+----------
       1 | 2012-03-11 18:30:00.000000+0000 |              79.9 |     Asha

(1 rows)
cqlsh:students> SELECT Roll_No, StudName FROM Students_Info LIMIT 2;

 roll_no | studname
---------+----------
       5 |    Rohan
       1 |     Asha

(2 rows)
cqlsh:students> SELECT Roll_No AS USN FROM Students_Info;

 usn
-----
   5
   1
   2
   4
   3

(5 rows)
cqlsh:students> UPDATE Students_Info
            ... SET StudName = 'David Sheen'
            ... WHERE Roll_No = 2;
cqlsh:students> UPDATE Students_Info SET Roll_No = 6 WHERE Roll_No = 3; -- ❌ ERROR!
InvalidRequest: Error from server: code=2200 [Invalid query] message="PRIMARY KEY part roll_no found in SET part"
```

```
cqlsh:students> DELETE Last_Exam_Percent FROM Students_Info WHERE Roll_No = 2;
cqlsh:students> DELETE FROM Students_Info WHERE Roll_No = 2;
cqlsh:students> ALTER TABLE Students_Info ADD hobbies SET<text>;
cqlsh:students> ALTER TABLE Students_Info ADD languages LIST<text>;
cqlsh:students> UPDATE Students_Info
            ... SET hobbies = hobbies + {'Chess', 'Table Tennis'}
            ... WHERE Roll_No = 1;
cqlsh:students> CREATE TABLE library_book (
            ...     counter_value counter,
            ...     book_name text,
            ...     stud_name text,
            ...     PRIMARY KEY(book_name, stud_name)
            ... );
cqlsh:students> UPDATE library_book
            ... SET counter_value = counter_value + 1
            ... WHERE book_name = 'Big Data Analytics' AND stud_name = 'Jeet';
cqlsh:students> CREATE TABLE userlogin (
            ...     userid int PRIMARY KEY,
            ...     password text
            ... );
cqlsh:students> INSERT INTO userlogin (userid, password)
            ... VALUES (1, 'infy') USING TTL 30;
cqlsh:students> SELECT TTL(password) FROM userlogin WHERE userid = 1;

 ttl(password)
---------------
            20

(1 rows)
cqlsh:students> COPY Students_Info TO '/home/bmscecse/Desktop/Student_Info.csv';
Using 16 child processes

Starting copy of students.students_info with columns [roll_no, dateofjoining, hobbies, languages, last_exam_percent, studname].
Processed: 4 rows; Rate:      38 rows/s; Avg. rate:      38 rows/s
4 rows exported to 1 files in 0.124 seconds.
cqlsh:students> COPY Students_Info FROM '/home/bmscecse/Desktop/Student_Info.csv';
Using 16 child processes

Starting copy of students.students_info with columns [roll_no, dateofjoining, hobbies, languages, last_exam_percent, studname].
Processed: 4 rows; Rate:       7 rows/s; Avg. rate:      11 rows/s
4 rows imported from 1 files in 0.377 seconds (0 skipped).
cqlsh:students> COPY person (id, fname, lname) FROM STDIN;
Column family person not found
cqlsh:students> COPY Students_Info TO STDOUT;
5,2012-03-11 18:30:00.000+0000,,,56.9,Rohan
1,2012-03-11 18:30:00.000+0000,"{'Chess', 'Table Tennis'}",,79.9,Asha
4,2012-03-11 18:30:00.000+0000,,,67.9,Smith
3,2012-03-11 18:30:00.000+0000,,,90.9,Shanthi
cqlsh:students>
```

PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

Command with output:

```
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 7043.  Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 7227.  Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
bmscecse-HP-Elite-Tower-800-G9-Desktop-PC: secondarynamenode is running as process 7521.  Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 7808.  Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 7969.  Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab06
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano file1.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -mkdir /rgs
mkdir: `/rgs': File exists
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /abc
mkdir: `/abc': File exists
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 5 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 14:27 /Lab06
drwxr-xr-x   - hadoop supergroup          0 2024-05-14 14:45 /abc
drwxr-xr-x   - hadoop supergroup          0 2024-05-14 14:46 /hadoop_lab
drwxr-xr-x   - hadoop supergroup          0 2024-05-21 14:56 /output
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 14:34 /rgs
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -put /home/hadoop/Desktop/welcome.txt /abc/WC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/welcome.txt /abc/WC.t
copyFromLocal: `/abc/WC.txt': File exists
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -get /abc/WC.txt /home/hadoop/Downloads/WWC.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -getmerge /abc/ /home/hadoop/Desktop/Merge.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /abc/
# file: /abc
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x

hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyToLocal /abc/WC.txt /home/hadoop/Desktop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /abc/WC.txt
hadoop is an open source platform
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /abc/FFF /FFF
mv: `/abc/FFF': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /FFF
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /abc /FFF
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /CSE/ /LLL
cp: `/CSE/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /abc/ /LLL
cp: `/abc/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /FFF/ /LLL
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

PROGRAM – 5

Implement Wordcount program on Hadoop framework

Code & command with output:

Driver Code:

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}
```

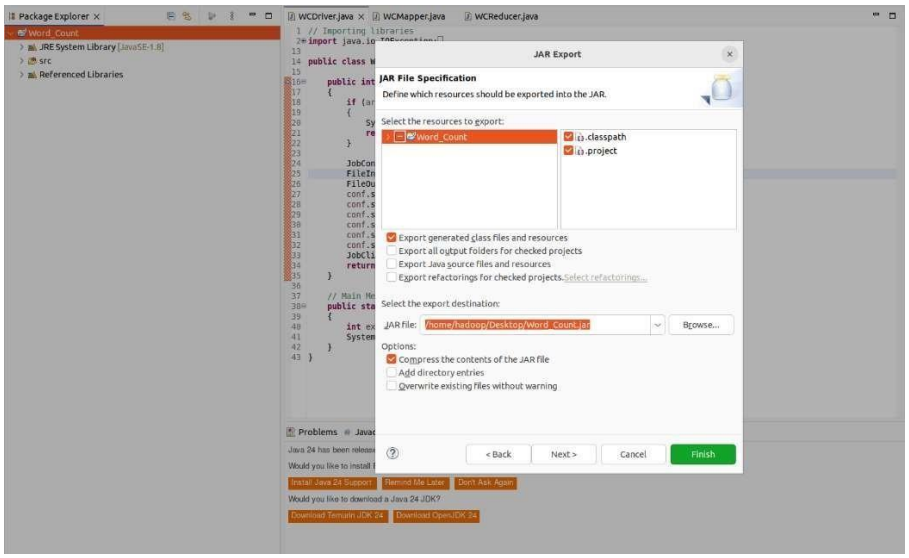
Mapper Code:

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
```

```
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}
```

Reducer Code:

```
// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
                    OutputCollector<Text, IntWritable> output,
                    Reporter reporter) throws IOException {
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}
```




```
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 7043.  Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 7227.  Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
bmscecse-HP-Elite-Tower-800-G9-Desktop-PC: secondarynamenode is running as process 7521.  Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 7808.  Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 7969.  Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hdfs dfs -mkdir /Lab06
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ nano file1.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -mkdir /rgs
mkdir: `/rgs': File exists
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -copyFromLocal -f /home/hadoop/Desktop/file1.txt /rgs/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/WordCount.jar wordcount.WordCount /rgs/test.txt /output
JAR does not exist or is not a normal file: /home/hadoop/Desktop/WordCount.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/WordCount.jar wordcount.WordCount /rgs/test.txt /output
JAR does not exist or is not a normal file: /home/hadoop/Desktop/WordCount.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/WordCount.jar wordcount.Word_Count /rgs/test.txt /output
JAR does not exist or is not a normal file: /home/hadoop/Desktop/WordCount.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop jar /home/hadoop/Desktop/Word_Count.jar wordcount.WordCount /rgs/test.txt /output
Exception in thread "main" java.lang.ClassNotFoundException: wordcount.WordCount
        at java.base/java.net.URLClassLoader.findClass(URLClassLoader.java:476)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:594)
        at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:527)
        at java.base/java.lang.Class.forName0(Native Method)
        at java.base/java.lang.Class.forName(Class.java:398)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:321)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -cat /output/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$ hadoop fs -ls /output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2024-05-21 14:56 /output/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2024-05-21 14:56 /output/part-00000
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~/Desktop$
```

PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all
a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

Code& command with output:

Driver Code:

```
package temp;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");

        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);

        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code:

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
}
```

Reducer Code:

```
package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;

        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
```

```
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
        }
    }
}
```

Name	Size	Type	Modified
META-INF	25 bytes	Folder	
.classpath	2.2 kB	unknown	06 May 2025, 14:40
.project	377 bytes	unknown	06 May 2025, 14:34
AverageDriver.class	1.6 kB	Java class	06 May 2025, 14:42
AverageMapper.class	2.4 kB	Java class	06 May 2025, 14:42
AverageReducer.class	2.3 kB	Java class	06 May 2025, 14:42

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ jps
7056 DataNode
7332 SecondaryNameNode
7638 ResourceManager
8231 Jps
5883 org.eclipse.equinox.launcher_1.6.1000.v20250227-1734.jar
7804 NodeManager
6877 NameNode
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /\
> ^C
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 4 items
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:00 /FFF
drwxr-xr-x   - hadoop supergroup          0 2025-04-15 15:34 /LLL
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:46 /file
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 15:18 /newDataFlair
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
ls: `/weather': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mkdir /weather
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -copyFromLocal /home/hadoop/Desktop/1901.txt /weather/test.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hadoop/Desktop/AverageTemperature.jar AverageDriver /weather/test.txt /weather/output
2025-05-06 14:59:23,239 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 14:59:23,279 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 14:59:23,340 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 14:59:23,393 INFO input.FileInputFormat: Total input files to process : 1
2025-05-06 14:59:23,422 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 14:59:23,487 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local91822813_0001
2025-05-06 14:59:23,488 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 14:59:23,560 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 14:59:23,560 INFO mapreduce.Job: Running job: job_local91822813_0001
2025-05-06 14:59:23,561 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 14:59:23,564 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,565 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,565 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,565 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 14:59:23,602 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 14:59:23,603 INFO mapred.LocalJobRunner: Starting task: attempt_local91822813_0001_m_000000_0
2025-05-06 14:59:23,615 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 14:59:23,615 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 14:59:23,615 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 14:59:23,622 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-05-06 14:59:23,624 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/weather/test.txt:0+888190
2025-05-06 14:59:23,658 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-05-06 14:59:23,658 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 14:59:23,658 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 14:59:23,658 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 14:59:23,658 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 14:59:23,660 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
```

```
2025-05-06 14:59:24,581 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=153118
                FILE: Number of bytes written=1493804
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1776380
                HDFS: Number of bytes written=8
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=59076
                Map output materialized bytes=72210
                Input split bytes=103
                Combine input records=0
                Combine output records=0
                Reduce input groups=1
                Reduce shuffle bytes=72210
                Reduce input records=6564
                Reduce output records=1
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1266679808
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=8
```

```
                Bytes Written=8
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather
Found 2 items
drwxr-xr-x   - hadoop supergroup          0 2025-05-06 14:59 /weather/output
-rw-r--r--   1 hadoop supergroup     888190 2025-05-06 14:50 /weather/test.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /weather/output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2025-05-06 14:59 /weather/output/_SUCCESS
-rw-r--r--   1 hadoop supergroup          8 2025-05-06 14:59 /weather/output/part-r-00000
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cat /weather/output/part-r-00000
1901    46
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

Code& command with output :

Driver Code

```
package meanmax;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;


import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MeanMaxDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Mean and Max Temperature");

        job.setJarByClass(MeanMaxDriver.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        job.setMapperClass(MeanMaxMapper.class);
        job.setReducerClass(MeanMaxReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

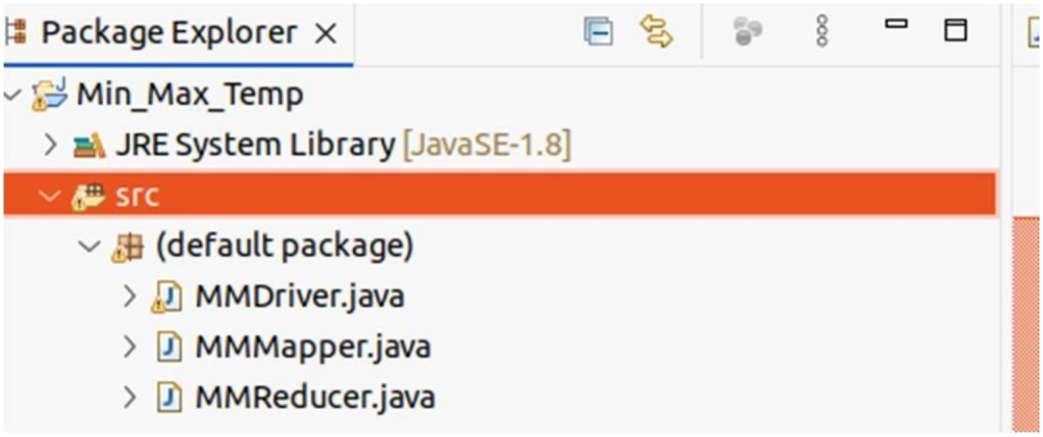
Mapper Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Mapper;

public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        String line = value.toString();

        // Extract month from positions 19-20
        String month = line.substring(19, 21);
        int temperature;
```

```
        // Extract temperature considering optional '+'
        if (line.charAt(87) == '+') {
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check
        String quality = line.substring(92, 93);

        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(month), new IntWritable(temperature));
        }
    }
}
```

Reducer Code

```
package meanmax;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context) throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
        int maxTemp = Integer.MIN_VALUE;

        for (IntWritable value : values) {
            int temp = value.get();
            sumTemp += temp;
            count++;

            if (temp > maxTemp) {
                maxTemp = temp;
            }
        }

        if (count > 0) {
            int avgTemp = sumTemp / count;
            String result = "mean=" + avgTemp + " max=" + maxTemp;
            context.write(key, new Text(result));
        }
    }
}
```




```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 5478.  Stop it first and ensure /tmp/hadoop-hadoop-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 5644.  Stop it first and ensure /tmp/hadoop-hadoop-datanode.pid file is empty before retry.
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
bmscecse-HP-Elite-Tower-800-G9-Desktop-PC: secondarynamenode is running as process 5931.  Stop it first and ensure /tmp/hadoop-hadoop-secondarynamenode.pid file is empty before retry.
Starting resourcemanager
resourcemanager is running as process 6214.  Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 6375.  Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/temp
copyFromLocal: `/rgs/temp': File exists
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir -p /rgs
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/1901 /rgs/1903
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hduser/Downloads/Mean_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Mean_Max_Temp.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hduser/Downloads/Min_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
JAR does not exist or is not a normal file: /home/hduser/Downloads/Min_Max_Temp.jar
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMDriver /rgs/avtemp.txt /out8
2025-05-06 15:23:05,430 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:23:05,471 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:23:05,471 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:23:05,531 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:23:05,575 INFO mapreduce.JobSubmitter: Cleaning up the staging area file:/tmp/hadoop/mapred/staging/hadoop1762005270/.staging/job_local1762005270_0001
Exception in thread "main" org.apache.hadoop.mapreduce.lib.input.InvalidInputException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:340)
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.listStatus(FileInputFormat.java:279)
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.getSplits(FileInputFormat.java:404)
        at org.apache.hadoop.mapreduce.JobSubmitter.writeNewSplits(JobSubmitter.java:310)
        at org.apache.hadoop.mapreduce.JobSubmitter.writeSplits(JobSubmitter.java:327)
        at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:200)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1678)
        at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
        at java.base/java.security.AccessController.doPrivileged(Native Method)
        at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
        at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
        at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
        at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
        at MMDriver.main(MMDriver.java:40)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
        at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
        at java.base/java.lang.reflect.Method.invoke(Method.java:566)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:313)
        ... 19 more
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /out8/*
cat: `/out8/*': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ ^C
```

```
Caused by: java.io.IOException: Input path does not exist: hdfs://localhost:9000/rgs/avtemp.txt
        at org.apache.hadoop.mapreduce.lib.input.FileInputFormat.singleThreadedListStatus(FileInputFormat.java:313)
        ... 19 more
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop jar /home/hadoop/Desktop/Min_Max_Temp.jar MMDriver /rgs/1903 /out8
2025-05-06 15:26:34,876 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-05-06 15:26:34,916 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-05-06 15:26:34,916 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-05-06 15:26:34,976 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-05-06 15:26:35,029 INFO input.FileInputFormat: Total input files to process : 1
2025-05-06 15:26:35,081 INFO mapreduce.JobSubmitter: number of splits:1
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1063792118_0001
2025-05-06 15:26:35,148 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-05-06 15:26:35,215 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2025-05-06 15:26:35,216 INFO mapreduce.Job: Running job: job_local1063792118_0001
2025-05-06 15:26:35,216 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2025-05-06 15:26:35,220 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,221 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,221 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2025-05-06 15:26:35,267 INFO mapred.LocalJobRunner: Waiting for map tasks
2025-05-06 15:26:35,268 INFO mapred.LocalJobRunner: Starting task: attempt_local1063792118_0001_m_000000_0
2025-05-06 15:26:35,278 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2025-05-06 15:26:35,278 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2025-05-06 15:26:35,279 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2025-05-06 15:26:35,286 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2025-05-06 15:26:35,287 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/rgs/1903:0+888190
2025-05-06 15:26:35,322 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2025-05-06 15:26:35,322 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2025-05-06 15:26:35,322 INFO mapred.MapTask: soft limit at 83886080
2025-05-06 15:26:35,322 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2025-05-06 15:26:35,322 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2025-05-06 15:26:35,325 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2025-05-06 15:26:35,440 INFO mapred.LocalJobRunner:
2025-05-06 15:26:35,441 INFO mapred.MapTask: Starting flush of map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: Spilling map output
2025-05-06 15:26:35,441 INFO mapred.MapTask: bufstart = 0; bufend = 45948; bufvoid = 104857600
2025-05-06 15:26:35,441 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26188144(104752576); length = 26253/6553600
2025-05-06 15:26:35,451 INFO mapred.MapTask: Finished spill 0
2025-05-06 15:26:35,456 INFO mapred.Task: Task:attempt_local1063792118_0001_m_000000_0 is done. And is in the process of committing
2025-05-06 15:26:35,458 INFO mapred.LocalJobRunner: map
2025-05-06 15:26:35,458 INFO mapred.Task: Task 'attempt_local1063792118_0001_m_000000_0' done.
2025-05-06 15:26:35,461 INFO mapred.Task: Final Counters for attempt_local1063792118_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=4359
                FILE: Number of bytes written=703803
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=888190
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
```

```
2025-05-06 15:26:36,222 INFO mapreduce.Job: Job job_local1063792118_0001 completed successfully
2025-05-06 15:26:36,233 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=126914
                FILE: Number of bytes written=1466688
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=1776380
                HDFS: Number of bytes written=74
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=6565
                Map output records=6564
                Map output bytes=45948
                Map output materialized bytes=59082
                Input split bytes=95
                Combine input records=0
                Combine output records=0
                Reduce input groups=12
                Reduce shuffle bytes=59082
                Reduce input records=6564
                Reduce output records=12
                Spilled Records=13128
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1052770304
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=888190
        File Output Format Counters
                Bytes Written=74
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /out8/*
01      4
02      0
03      7
04      44
05      100
06      168
07      219
08      198
09      141
10      100
11      19
12      3
```

PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

Code, command with output:

Driver Code:

```
package samples.topn;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code:

```
package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);
    private Text word = new Text();
    // characters to normalize into spaces
    private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
                        .toLowerCase()
                        .replaceAll(tokens, " ");
        StringTokenizer itr = new StringTokenizer(clean);
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}
```

Mapper Code:

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper
    extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");
        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));
            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code:

```
package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class WordCountReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Reducer Code :

```
package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer
    extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();
        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
        for (Text w : values) {
            words.add(w.toString());
        }
        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
        throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;
        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
```

```
            int cnt = entry.getKey();
            for (String w : entry.getValue()) {
                topList.add(new WordCount(w, cnt));
                seen++;
                if (seen == 10) break;
            }
            if (seen == 10) break;
        }

        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical order
        for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }

    // helper class
    private static class WordCount {
        String word;
        int count;
        WordCount(String w, int c) { word = w; count = c; }
    }
}
```

```
2025-04-29 15:32:09,761 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager
 at /0.0.0.0:8032
2025-04-29 15:32:09,829 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager
 at /0.0.0.0:8032
2025-04-29 15:32:09,918 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not pe
rformed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2025-04-29 15:32:09,944 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/h
adoop-yarn/staging/hadoop/.staging/job_1745919848818_0003
2025-04-29 15:32:10,138 INFO mapred.FileInputFormat: Total input files to process : 1
2025-04-29 15:32:10,227 INFO mapreduce.JobSubmitter: number of splits:2
2025-04-29 15:32:10,318 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1745919848818_000
3
2025-04-29 15:32:10,318 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-04-29 15:32:10,405 INFO conf.Configuration: resource-types.xml not found
2025-04-29 15:32:10,405 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-04-29 15:32:10,556 INFO impl.YarnClientImpl: Submitted application application_1745919848818_000
3
2025-04-29 15:32:10,574 INFO mapreduce.Job: The url to track the job: http://bmscecse-HP-Elite-Tower-
800-G9-Desktop-PC:8088/proxy/application_1745919848818_0003/
2025-04-29 15:32:10,575 INFO mapreduce.Job: Running job: job_1745919848818_0003
2025-04-29 15:32:15,652 INFO mapreduce.Job: Job job_1745919848818_0003 running in uber mode : false
2025-04-29 15:32:15,654 INFO mapreduce.Job:  map 0% reduce 0%
2025-04-29 15:32:18,772 INFO mapreduce.Job:  map 100% reduce 0%
2025-04-29 15:32:22,799 INFO mapreduce.Job:  map 100% reduce 100%
2025-04-29 15:32:23,824 INFO mapreduce.Job: Job job_1745919848818_0003 completed successfully
2025-04-29 15:32:23,882 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=215
                FILE: Number of bytes written=829242
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=306
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=11
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=2555
                Total time spent by all reduces in occupied slots (ms)=1281
                Total time spent by all map tasks (ms)=2555
                Total time spent by all reduce tasks (ms)=1281
                Total vcore-milliseconds taken by all map tasks=2555
                Total vcore-milliseconds taken by all reduce tasks=1281
                Total megabyte-milliseconds taken by all map tasks=2616320
                Total megabyte-milliseconds taken by all reduce tasks=1311744
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /rgs/output
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2025-04-29 15:32 /rgs/output/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2025-04-29 15:32 /rgs/output/part-00000
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cat /rgs/output/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

Code, command with output:

```
bmscecse@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ spark-shell
25/05/20 11:28:13 WARN Utils: Your hostname, bmscecse-HP-Elite-Tower-600-G9-Desktop-PC resolves to a loopback address: 127.0.1.1; using 10.124.3.80 instead (on interface eno1)
25/05/20 11:28:13 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.spark.unsafe.Platform (file:/opt/spark/jars/spark-unsafe_2.12-3.0.3.jar) to constructor java.nio.DirectByteBuffer(long,int)
WARNING: Please consider reporting this to the maintainers of org.apache.spark.unsafe.Platform
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
25/05/20 11:28:13 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Spark context Web UI available at http://10.124.3.80:4040
Spark context available as 'sc' (master = local[*], app id = local-1747720695950).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.0.3
      /_/

Using Scala version 2.12.10 (OpenJDK 64-Bit Server VM, Java 11.0.26)
Type in expressions to have them evaluated.
Type :help for more information.

scala> for (i <- 1 to 100) print(i + " ")
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 6
5 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
```

PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

Code, command with output:

```
scala> val rdd = spark.sparkContext.textFile("file:/home/bmscecse/Desktop/scala")
rdd: org.apache.spark.rdd.RDD[String] = file:/home/bmscecse/Desktop/scala MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = rdd.flatMap(_.split("\\s+")).map(word => (word.toLowerCase, 1)).reduceByKey(_ + _).filter(_._2 > 4)
counts: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[5] at filter at <console>:25

scala> counts.collect().foreach{ case (word, count) => println(s"$word $count") }
spark 6

scala>
```

PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

Code, command with output:

```
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

df = spark.createDataFrame(lines, "string").toDF("value")
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.2.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.5.0)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
+------+
|word  |
+------+
|hello |
|hate  |
|hate  |
|love  |
|dont  |
|want  |
|cant  |
|put   |
|nobody|
|else  |
+------+
```